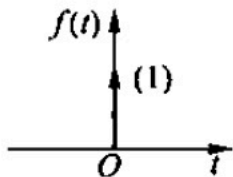
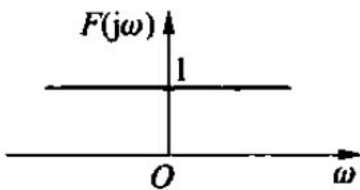
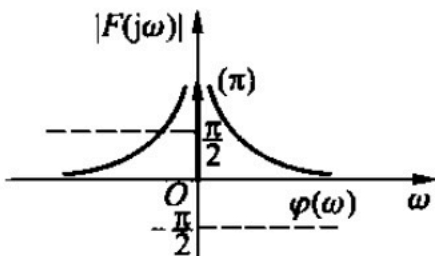
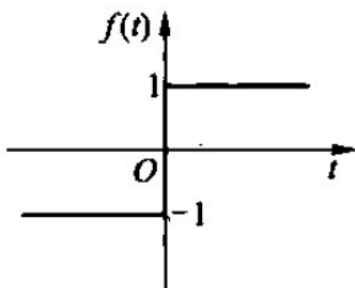
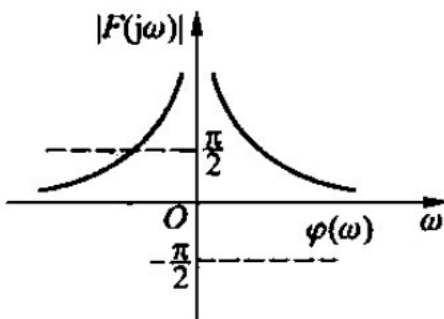
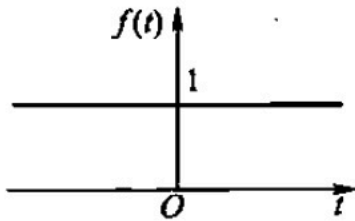
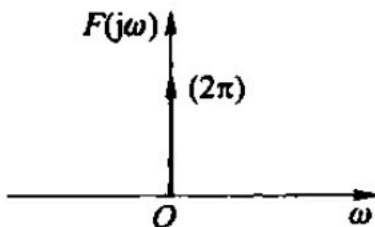
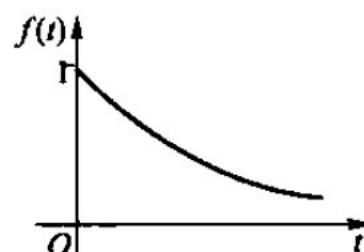
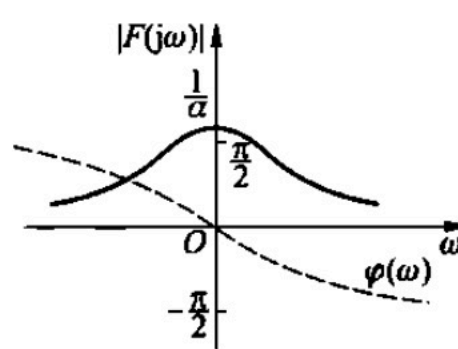
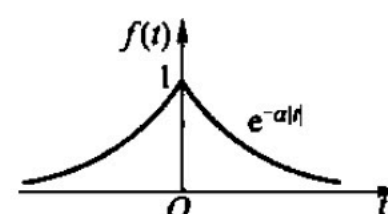
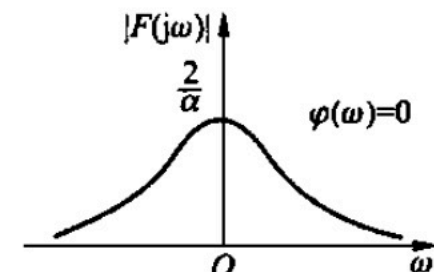
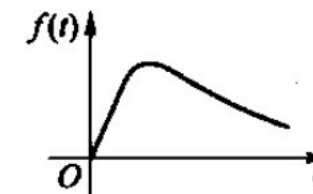
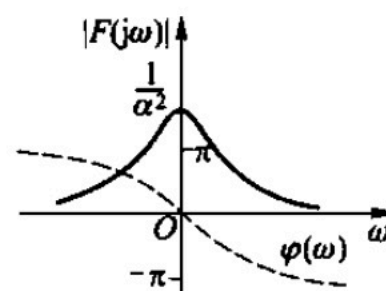
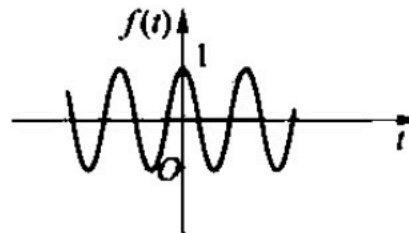
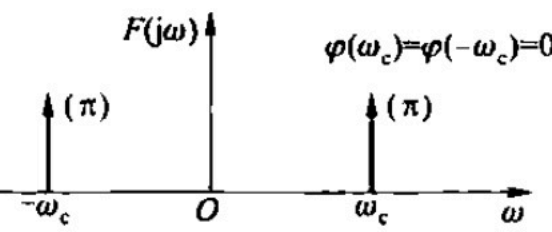
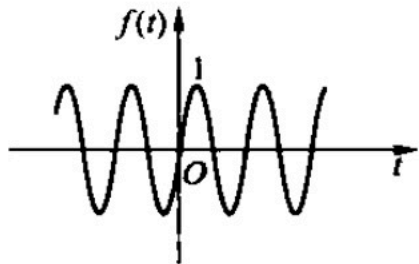
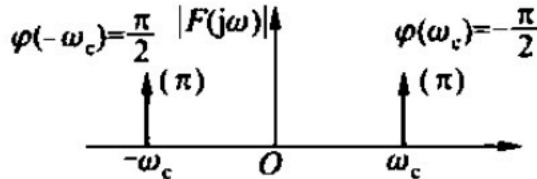
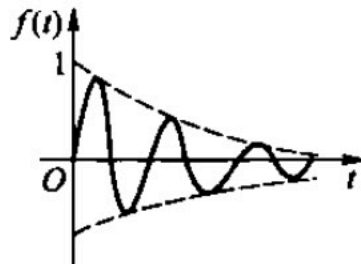
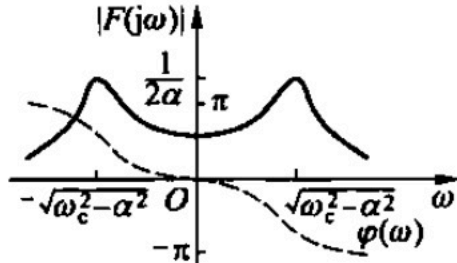
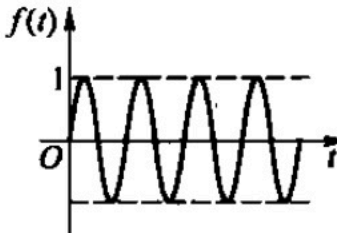
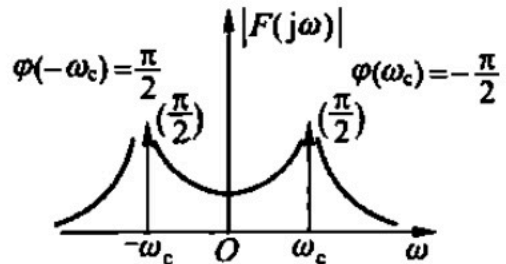
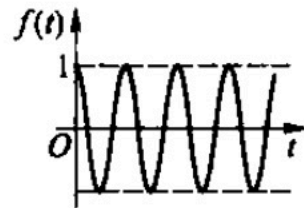
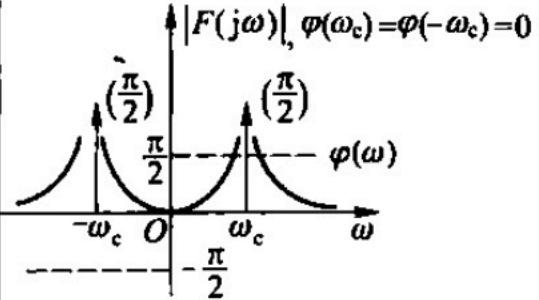


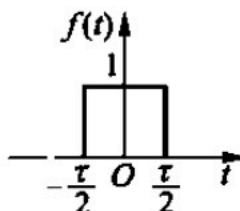
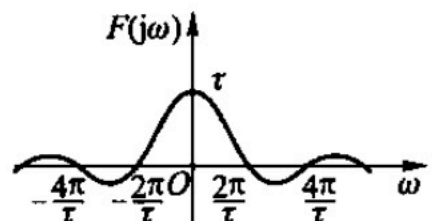
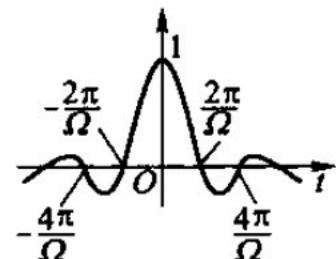
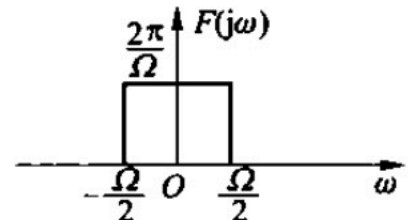
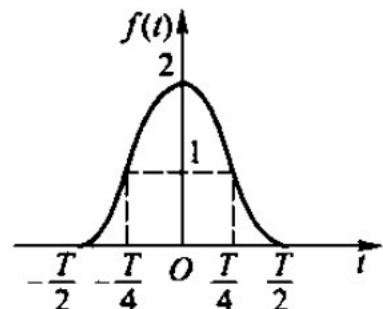
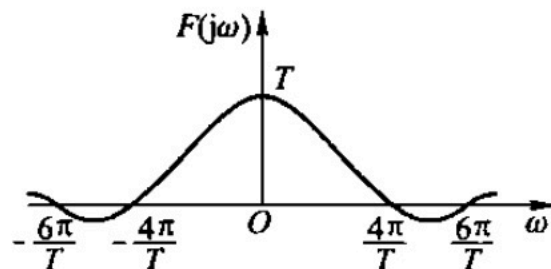
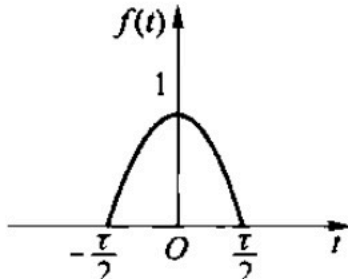
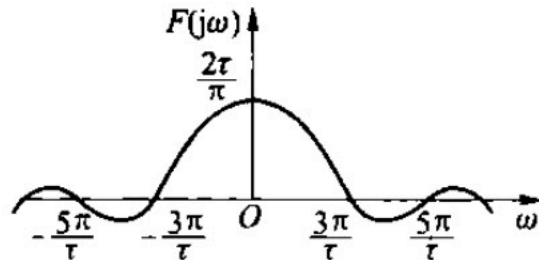
表 3-1 傅里叶变换表 (几种常用函数及其频谱)

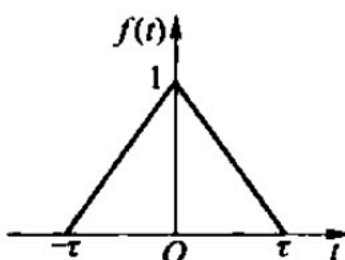
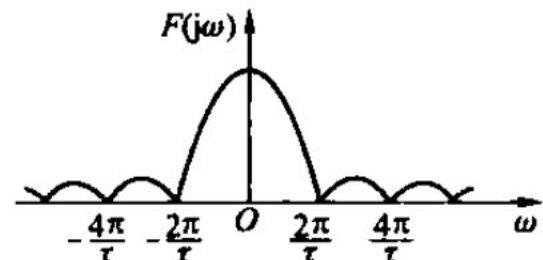
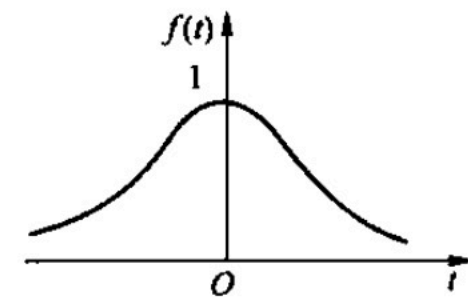
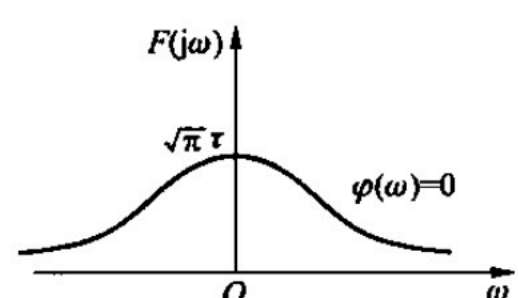
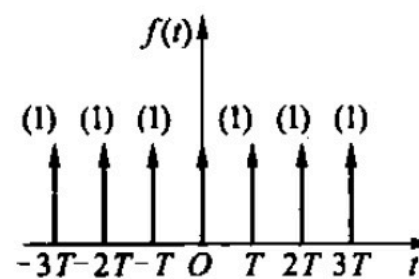
序号	名称	时间函数 $f(t)$		频谱函数 $F(j\omega) = F(j\omega) e^{j\varphi(\omega)}$	
1	单位冲激	$\delta(t)$		1	
2	单位阶跃	$\varepsilon(t)$		$\pi\delta(\omega) + \frac{1}{j\omega}$	
3	符号函数	$\text{sgn } t = \varepsilon(t) - \varepsilon(-t)$		$\frac{2}{j\omega}$	
4	单位直流	1		$2\pi\delta(\omega)$	

续表

序号	名称	时间函数 $f(t)$		频谱函数 $F(j\omega) = F(j\omega) e^{j\varphi(\omega)}$	
5	单边指数函数	$e^{-\alpha t}\varepsilon(t)$		$\frac{1}{\alpha + j\omega}$	
6	双边指数函数	$e^{-\alpha t }$		$\frac{2\alpha}{\alpha^2 + \omega^2}$	
7	指数脉冲	$te^{-\alpha t}\varepsilon(t)$		$\frac{1}{(\alpha + j\omega)^2}$	
8	单位余弦	$\cos \omega_c t$		$\pi[\delta(\omega + \omega_c) + \delta(\omega - \omega_c)]$	

序号	名称	时间函数 $f(t)$	频谱函数 $F(j\omega) = F(j\omega) e^{j\varphi(\omega)}$
9	单位正弦	$\sin \omega_c t$	 $j\pi[\delta(\omega + \omega_c) - \delta(\omega - \omega_c)]$ 
10	减幅正弦	$e^{-\alpha t} \sin \omega_c t \varepsilon(t)$	 $\frac{\omega_c}{(\alpha + j\omega)^2 + \omega_c^2}$ 
11	阶跃正弦	$\sin \omega_c t \varepsilon(t)$	 $\frac{\pi}{2j} [\delta(\omega - \omega_c) - \delta(\omega + \omega_c)] + \frac{\omega_c}{\omega_c^2 - \omega^2}$ 
12	阶跃余弦	$\cos \omega_c t \varepsilon(t)$	 $\frac{\pi}{2} [\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] + \frac{j\omega}{\omega_c^2 - \omega^2}$ 

序号	名称	时间函数 $f(t)$	频谱函数 $F(j\omega) = F(j\omega) e^{j\varphi(\omega)}$
13	矩形脉冲 (门函数)	$G_{\tau}(t) = \varepsilon\left(t + \frac{\tau}{2}\right) - \varepsilon\left(t - \frac{\tau}{2}\right)$ 	$\tau \text{Sa}\left(\frac{\tau\omega}{2}\right) = \tau \frac{\sin(\tau\omega/2)}{\tau\omega/2}$ 
14	抽样函数	$\text{Sa}\left(\frac{\Omega t}{2}\right) = \frac{\sin(\Omega t/2)}{\Omega t/2}$ 	$G_{\Omega}(\omega) = \frac{2\pi}{\Omega} \left[\varepsilon\left(\omega + \frac{\Omega}{2}\right) - \varepsilon\left(\omega - \frac{\Omega}{2}\right) \right]$ 
15	升余弦脉冲	$\left[1 + \cos\left(\frac{2\pi}{T}t\right) \right] \left[\varepsilon\left(t + \frac{T}{2}\right) - \varepsilon\left(t - \frac{T}{2}\right) \right]$ 	$\frac{8\pi^2 \sin(T\omega/2)}{\omega(4\pi^2 - T^2\omega^2)}$ 
16	半余弦脉冲	$\cos\left(\frac{\pi}{\tau}t\right) \left[\varepsilon\left(t + \frac{\tau}{2}\right) - \varepsilon\left(t - \frac{\tau}{2}\right) \right]$ 	$\frac{2\tau}{\pi} \cdot \frac{\cos(\tau\omega/2)}{1 - \tau^2\omega^2/\pi^2}$ 

序号	名称	时间函数 $f(t)$		频谱函数 $F(j\omega) = F(j\omega) e^{j\varphi(\omega)}$	
17	三角形脉冲	$f(t) = 1 - \frac{ t }{\tau}, \text{ 当 } t < \tau$ $f(t) = 0, \text{ 当 } t > \tau$		$\tau \left[\text{Sa} \left(\frac{\omega\tau}{2} \right) \right]^2 =$ $\tau \left[\frac{\sin(\omega\tau/2)}{\omega\tau/2} \right]^2$	
18	高斯脉冲	$e^{-\left(\frac{t}{\tau}\right)^2}$		$\sqrt{\pi} \tau e^{-\left(\frac{\omega\tau}{2}\right)^2}$	
19	冲激序列	$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT)$		$\Omega \delta_\Omega(\omega)$ $= \Omega \sum_{n=-\infty}^{\infty} \delta(\omega - n\Omega)$ $\Omega = \frac{2\pi}{T}$	